

Big line or big convex polygon: the dependence on ℓ is uniformly linear

github.com/graph-theory-AI

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The argument was found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2405.03455__00; original writeup in `attacks/2405.03455__00/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every identity and inequality by hand and re-checked them symbolically and numerically (all $n \leq 20000$ and $n = 10^5, 10^6$ for the asymptotic step), fetched the TeX source of the source paper to confirm the two theorems used and the exact wording of the open problem, and read the only citing paper (Furukawa, arXiv:2501.03645), which proves a strictly weaker bound; its verdict was **CONFIRMED**. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in [Appendix A](#); the constant C'_0 of the writeup is explained as the rescaling of C'_0 by the base of the logarithm; and the referee's clarifications (the explicit handling of the ceiling, the edge cases $\ell = 3$ and $k = n = 3$, the numerical value of the constant, the comparison with Furukawa's bound) were added as remarks. No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

Let $ES_\ell(n)$ be the least N such that every N -point set in the plane contains ℓ collinear points or n points in convex position. Conlon, Fox, He, Mubayi, Suk and Verstraete proved $(3\ell - 1)2^{n-5} < ES_\ell(n) < \ell^2 2^{n+C\sqrt{n\log n}}$ for all $\ell, n \geq 3$ and asked for the correct dependence on ℓ . We show that there is an absolute constant C_1 with

$$ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}} \quad (\ell, n \geq 3),$$

so that $ES_\ell(n)/\ell$ lies between $2^n/12$ and $2^{n+C_1\sqrt{n\log n}}$ uniformly in both parameters: the dependence on ℓ is linear. The proof is a two-step bootstrap. An elementary alteration lemma extracts from any N -point set with fewer than ℓ collinear points a subset of size $\frac{1}{4}(N/\ell)^{(k-2)/(k-1)}$ with no k collinear points; applied with $k = n$ and combined with the diagonal case $ES_n(n)$ of the known upper bound, the exponent loss $(n-1)/(n-2)$ costs only a factor absorbed by the subexponential term.

1 Introduction

For integers $\ell, n \geq 3$ let $ES_\ell(n)$ denote the minimum N such that every set of N points in the plane contains either ℓ collinear points or n points in convex position (the vertices of a convex n -gon). For $\ell = 3$ this is the classical Erdős–Szekeres number $ES(n)$. Conlon, Fox, He, Mubayi, Suk and Verstraete [1] proved that there is a constant $C > 0$ such that for all $\ell, n \geq 3$

$$(3\ell - 1)2^{n-5} + 1 \leq ES_\ell(n) \leq \ell^2 2^{n+C\sqrt{n\log n}} \quad (1)$$

(their Theorems 1.2 and 1.1, respectively), and at the end of their introduction they write:

Problem 1.1 ([1]). “It remains an interesting open problem to determine the correct dependence of $ES_\ell(n)$ on ℓ .”

The lower bound in (1) is linear in ℓ and the upper bound quadratic. In the regime where n is fixed and $\ell \rightarrow \infty$ the same paper’s cups–caps theorem gives $ES_\ell(n) = O(\ell)$, but the joint dependence was open. We show that the upper bound can be made linear in ℓ uniformly, with no restriction on the relative sizes of ℓ and n .

Theorem 1.2 (Main theorem). *There is an absolute constant $C_1 > 0$ such that for all integers $\ell, n \geq 3$*

$$ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}}.$$

Together with the lower bound in (1) this gives, for all $\ell, n \geq 3$,

$$\frac{1}{12} \ell 2^n < ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}}.$$

Thus the dependence on ℓ is uniformly linear; the remaining gap, a factor $12 \cdot 2^{C_1\sqrt{n\log n}}$, depends on n alone and is the same gap that remains open for the classical Erdős–Szekeres number itself (see [Remark 3.1](#)).

Idea of proof. The whole argument is a transfer from the diagonal case. An alteration lemma ([Lemma 2.1](#)) shows that an N -point set with fewer than ℓ collinear points contains a subset of size at least $\frac{1}{4}(N/\ell)^{(k-2)/(k-1)}$ with no k collinear points; the point is that the collinear k -subsets form a k -uniform hypergraph with fewer than $N^2\ell^{k-2}$ edges, because every pair of points determines a line and a line with $s < \ell$ points carries only $\binom{s}{2}\ell^{k-2}$ or fewer such k -subsets. Feeding a set with no k collinear points and at least $ES_k(n)$ points into the definition of $ES_k(n)$ produces n points in convex position, which yields the transfer inequality $ES_\ell(n) \leq \lceil \ell(4ES_k(n))^{(k-1)/(k-2)} \rceil$ ([Theorem 2.2](#)). Choosing $k = n$ and inserting the source’s bound $ES_n(n) < n^2 2^{n+C_0\sqrt{n\log n}}$, the exponent $(n-1)/(n-2) = 1 + 1/(n-2)$ multiplies the exponent $n + O(\sqrt{n\log n})$ by a factor whose excess $1/(n-2)$ contributes only $O(1)$, so everything beyond $\ell 2^n$ is absorbed into $2^{O(\sqrt{n\log n})}$. [\[full proof\]](#)

2 The transfer inequality

Lemma 2.1 (Alteration lemma). *Let P be a set of N points in the plane with fewer than ℓ points on every line, and let $k \geq 3$ be an integer. Then P contains a subset Q with no k collinear points and*

$$|Q| \geq \frac{1}{4} \left(\frac{N}{\ell} \right)^{(k-2)/(k-1)}.$$

Idea of proof. Let H be the k -uniform hypergraph on P whose edges are the collinear k -subsets, with e edges. Since every pair of points lies on a unique line and each line carries at most $\ell - 1$ points, $e < N^2\ell^{k-2}$. If $e < N/2$, deleting one point per edge leaves more than $N/2$ points. Otherwise keep each point independently with probability $p = (N/2e)^{1/(k-1)}$, so that the expected number of kept points minus the expected number of surviving edges is exactly $pN/2$; delete one point from each surviving edge. In both cases the independent set obtained has the claimed size. [\[full proof\]](#)

Theorem 2.2 (Transfer inequality). *For all integers $k, \ell, n \geq 3$,*

$$ES_\ell(n) \leq \left\lceil \ell (4ES_k(n))^{(k-1)/(k-2)} \right\rceil.$$

Idea of proof. Let $M = ES_k(n)$ and $N = \lceil \ell(4M)^{(k-1)/(k-2)} \rceil$. An N -point set without ℓ collinear points contains, by [Lemma 2.1](#), a subset Q with no k collinear points and $|Q| \geq M$; by the definition of $ES_k(n)$, Q contains n points in convex position. [\[full proof\]](#)

3 Remarks

Remark 3.1 (What remains open). The argument does not narrow the factor $2^{O(\sqrt{n \log n})}$ between the lower and upper bounds as functions of n ; it removes the extra factor ℓ from the upper bound of [1] without worsening the n -dependence beyond the absolute constant in the subexponential term. Read maximally strictly, “determine the correct dependence on ℓ ” might ask for the exact coefficient of ℓ for each n ; that reading subsumes the classical Erdős–Szekeres problem ($2^{n-2} + 1$ versus $2^{n+O(\sqrt{n \log n})}$) and is not addressed.

Remark 3.2 (Constants and the ceiling). The referee (Appendix B) made the last step quantitative: writing the source bound as $ES_n(n) < n^2 2^{n+C'_0 \sqrt{n \log_2 n}}$, the argument of the ceiling in Theorem 2.2 with $k = n$ is at least 12, so $\lceil x \rceil \leq \frac{13}{12}x$ there, and the inequality $\frac{n-1}{n-2} \log_2(4n^2 2^{n+C'_0 \sqrt{n \log_2 n}}) + \log_2 \frac{13}{12} \leq n + C_1 \sqrt{n \log_2 n}$ holds for all $n \geq 3$ with, for instance, $C_1 = 12.18$ when $C'_0 = 3$ (the worst case being $n = 3$). The constant $\frac{1}{12}$ in the lower bound is tight only at $\ell = 3$ and is inherited from the construction of [1].

Remark 3.3 (Edge cases). For $\ell = 3$ no line contains $k \geq 3$ points, the hypergraph in Lemma 2.1 is empty and the lemma is vacuous but true. For $k = n = 3$ one has $ES_3(3) = 3$ and the transfer inequality gives $ES_\ell(3) \leq 144\ell$, wasteful but correct against the true value $ES_\ell(3) = \ell$. The strict inequality $\frac{1}{12}\ell 2^n < ES_\ell(n)$ at $\ell = 3$ uses the “+1” in the lower bound of (1).

Remark 3.4 (Related work and novelty). The only paper citing [1] found by the referee is Furukawa [2], who proves $ES_\ell(n) \leq (\ell - 3) \binom{ES(n)-1}{2} + ES(n)$, i.e. roughly $\ell 4^{n+O(\sqrt{n \log n})}$, by a maximal-general-position-subset argument, and remarks that this is weaker than (1) unless ℓ is much larger than n . The bound of Theorem 1.2 is stronger than both. The alteration lemma is a standard hypergraph independence bound; its case $k = 3$ is the classical step of the general-position subset selection literature (e.g. Payne and Wood [3]). The bootstrap through the diagonal value $ES_n(n)$, which is what makes the exponent loss harmless, was not found in the literature; given that the six authors of [1] left the question open and that the argument is short, the referee explicitly recommends a folklore check. Correctness does not depend on novelty.

Remark 3.5 (Computational checks). The referee verified symbolically the binomial identity and the two algebraic identities in the proof of Lemma 2.1 for $3 \leq k \leq 14$, the inequality $2^{-1-1/(k-1)} \geq \frac{1}{4}$ for $k < 1000$, the arithmetic of Theorem 2.2 on a grid of (k, M, ℓ) , and the asymptotic step of Theorem 1.2 for all $3 \leq n \leq 20000$ and $n = 10^5, 10^6$; it also stress-tested Lemma 2.1 on thirty concrete point sets (grids, unions of lines, clustered and random sets). Scripts are in `verification/scripts/2405.03455__00/`.

A Full proofs

Lemma 2.1 (Alteration lemma). *Let P be a set of N points in the plane with fewer than ℓ points on every line, and let $k \geq 3$ be an integer. Then P contains a subset Q with no k collinear points and*

$$|Q| \geq \frac{1}{4} \left(\frac{N}{\ell} \right)^{(k-2)/(k-1)}.$$

Proof. Form the k -uniform hypergraph H on vertex set P whose edges are the collinear k -subsets of P , and let $e = e(H)$. An independent set of H is exactly a subset of P with no k collinear points.

Counting edges. For every line L containing at least k points of P put $s_L = |P \cap L|$; by hypothesis $s_L \leq \ell - 1$. Every collinear k -subset lies on exactly one line, so $e = \sum_L \binom{s_L}{k}$. For $s \geq k$,

$$\binom{s}{k} = \binom{s}{2} \frac{\binom{s-2}{k-2}}{\binom{k-2}{2}} \leq \binom{s}{2} \ell^{k-2},$$

since $\binom{s-2}{k-2} \leq (s-2)^{k-2} \leq \ell^{k-2}$ and $\binom{k}{2} \geq 1$. Every pair of distinct points of P lies on a unique line, so $\sum_L \binom{s_L}{2} \leq \binom{N}{2}$. Hence

$$e \leq \ell^{k-2} \sum_L \binom{s_L}{2} \leq \ell^{k-2} \binom{N}{2} < N^2 \ell^{k-2}. \quad (2)$$

Put $a = (k-2)/(k-1) \in (0, 1)$.

Case $e < N/2$. Delete one vertex from each edge. At most $e < N/2$ vertices are deleted, so an independent set of size greater than $N/2$ remains, and $N/2 > N/4 \geq \frac{1}{4}(N/\ell)^a$ because $(N/\ell)^a \leq N^a \leq N$.

Case $e \geq N/2$. Put $p = (N/2e)^{1/(k-1)}$, so $0 < p \leq 1$. Choose each vertex of P independently with probability p ; let X be the number of chosen vertices and Y the number of edges all of whose k vertices are chosen. Then $\mathbb{E}X = pN$ and $\mathbb{E}Y = p^k e$, and since $p^{k-1}e = N/2$,

$$\mathbb{E}[X - Y] = pN - p^k e = pN - p \cdot \frac{N}{2} = \frac{pN}{2}.$$

Hence some outcome has $X - Y \geq pN/2$. In that outcome delete one chosen vertex from each of the Y edges contained in the chosen set; this destroys every such edge and leaves an independent set of size at least $X - Y \geq pN/2$. By (2),

$$\frac{pN}{2} = \frac{N}{2} \left(\frac{N}{2e} \right)^{1/(k-1)} \geq \frac{N}{2} \left(\frac{1}{2N\ell^{k-2}} \right)^{1/(k-1)} = 2^{-1-1/(k-1)} \left(\frac{N}{\ell} \right)^{(k-2)/(k-1)} \geq \frac{1}{4} \left(\frac{N}{\ell} \right)^{(k-2)/(k-1)},$$

where the middle equality is the identity $N \cdot N^{-1/(k-1)} = N^{(k-2)/(k-1)}$ together with $\ell^{-(k-2)/(k-1)}$, and the last inequality is $1 + 1/(k-1) \leq 2$ for $k \geq 2$ (for $k = 3$ the factor is $2^{-3/2} \approx 0.354 > \frac{1}{4}$). This proves the lemma. $\square$

Theorem 2.2 (Transfer inequality). *For all integers $k, \ell, n \geq 3$,*

$$ES_\ell(n) \leq \left\lceil \ell (4 ES_k(n))^{(k-1)/(k-2)} \right\rceil.$$

Proof. Let $M = ES_k(n)$, $r = (k-1)/(k-2)$ and $N = \lceil \ell(4M)^r \rceil$. Let P be any set of N points in the plane. If P contains ℓ collinear points there is nothing to prove. Otherwise every line contains fewer than ℓ points of P , and Lemma 2.1 gives $Q \subseteq P$ with no k collinear points and

$$|Q| \geq \frac{1}{4} \left(\frac{N}{\ell} \right)^{1/r} \geq \frac{1}{4} ((4M)^r)^{1/r} = M,$$

using $N/\ell \geq (4M)^r$. By the definition of $M = ES_k(n)$, every M -point subset of Q contains k collinear points or n points in convex position; the first alternative is impossible inside Q , so Q , and hence P , contains n points in convex position. Therefore every N -point set contains ℓ collinear points or n points in convex position, i.e. $ES_\ell(n) \leq N$. $\square$

Theorem 1.2 (Main theorem). *There is an absolute constant $C_1 > 0$ such that for all integers $\ell, n \geq 3$*

$$ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}}.$$

Together with the lower bound in (1) this gives, for all $\ell, n \geq 3$,

$$\frac{1}{12} \ell 2^n < ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}}.$$

Proof. Apply [Theorem 2.2](#) with $k = n$:

$$ES_\ell(n) \leq \left\lceil \ell (4 ES_n(n))^{(n-1)/(n-2)} \right\rceil. \quad (3)$$

Theorem 1.1 of [1] holds for all pairs $\ell, n \geq 3$; taking both parameters equal to n gives an absolute constant C_0 with

$$ES_n(n) < n^2 2^{n+C_0\sqrt{n\log n}}. \quad (4)$$

Let C'_0 be C_0 rescaled so that $2^{C_0\sqrt{n\log n}} = 2^{C'_0\sqrt{n\log_2 n}}$ (i.e. $C'_0 = C_0\sqrt{\ln 2}$ if $\log$ in (4) is the natural logarithm), and let $r = (n-1)/(n-2) = 1 + 1/(n-2)$. Taking base-2 logarithms in (4),

$$\begin{aligned} \log_2(4ES_n(n))^r &\leq r(n + 2\log_2 n + 2 + C'_0\sqrt{n\log_2 n}) \\ &= n + C'_0\sqrt{n\log_2 n} + 2\log_2 n + 2 + \frac{n + C'_0\sqrt{n\log_2 n} + 2\log_2 n + 2}{n-2}. \end{aligned}$$

For $n \geq 3$ every term after the leading n is $O(\sqrt{n\log n})$ with an absolute implied constant: $2\log_2 n + 2 = O(\sqrt{n\log n})$, and the last fraction is at most $\frac{n}{n-2} + \frac{C'_0\sqrt{n\log_2 n} + 2\log_2 n + 2}{n-2} \leq 3 + O(\sqrt{n\log n})$, while $\sqrt{n\log_2 n} \geq \sqrt{3\log_2 3} > 2$ for $n \geq 3$ so constants are absorbed as well. The ceiling in (3) changes the bound by a factor at most $\frac{13}{12}$ (its argument is at least $\ell \cdot 4ES_n(n) \geq 12$; cf. [Remark 3.2](#)), which is likewise absorbed. Hence there is an absolute C_1 with

$$ES_\ell(n) \leq \ell 2^{n+C_1\sqrt{n\log n}} \quad (\ell, n \geq 3).$$

For the lower bound, Theorem 1.2 of [1] gives $ES_\ell(n) \geq (3\ell-1)2^{n-5} + 1 > (3\ell-1)2^{n-5}$, and

$$(3\ell-1)2^{n-5} - \frac{1}{12}\ell 2^n = \frac{2^n(\ell-3)}{96} \geq 0 \quad (\ell \geq 3),$$

so $ES_\ell(n) > \frac{1}{12}\ell 2^n$. Combining the two bounds proves the theorem. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02/03) on the *original* writeup. Its step numbers (S1–S17) refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2405.03455__00/`. Treat it as machine output, not as an authority.

Target: `attacks/2405.03455__00/output.md` Problem: determine the correct dependence of $ES_l(n)$ on l , where $ES_l(n)$ is the minimum N such that every N -point planar set contains l collinear points or n points in convex position (Conlon–Fox–He–Mubayi–Suk–Verstraete, arXiv:2405.03455, published Comput. Geom. 131 (2026)). The source paper proves $(3l-1)2^{n-5} < ES_l(n) < l^2 2^{n+C\sqrt{n\log n}}$ and asks for the correct l -dependence.

The writeup claims: $ES_l(n) \leq l \cdot 2^{n+C_1\sqrt{n\log n}}$ for all $l, n \geq 3$, via the transfer inequality $ES_l(n) \leq \text{ceil}(l(4ES_k(n))^{(k-1)/(k-2)})$ with $k = n$, so that the dependence on l is uniformly linear, matching the source lower bound.

B.1 Interpretation

The catalog problem is exactly the sentence at the end of the source paper’s introduction: “It remains an interesting open problem to determine the correct dependence of $ES_l(n)$ on l .” I verified this sentence verbatim in the arXiv TeX source (line 127 of `main.tex`). The context makes clear the question

is whether the l^2 in the upper bound can be brought down to match the linear lower bound uniformly in both parameters (the regime n fixed, $l \rightarrow \infty$ was already known to be $\Theta(l)$ via the paper's cups-caps theorem, as the paper itself states).

The writeup's formalization is faithful: it uses the paper's exact definition of $ES_l(n)$ (at least l collinear, or n in convex position; arbitrary point sets), imposes no hidden restrictions on l or n , and proves a uniform upper bound $l \cdot 2^{n+C_1 \sqrt{n \log n}}$ valid for all $l, n \geq 3$. Combined with the source lower bound this pins $ES_l(n)/l$ between $2^{n/12}$ and $2^{n+C_1 \sqrt{n \log n}}$ uniformly, i.e., the dependence on l is linear with no restriction on the relative sizes of l and n . A reasonable author of the source paper would, in my judgment, consider this a resolution of the stated open problem about the l -dependence: the remaining gap, $2^{O(\sqrt{n \log n})}$, is a function of n only and is precisely the same gap that remains open for the classical Erdos-Szekeres number $ES(n)$ itself ($2^{n-2}+1$ vs $2^{n+O(\sqrt{n \log n})}$), so it cannot reasonably be charged to the l -dependence question. There is no interpretation gaming: the writeup attacks the intended reading and honestly flags what is not resolved (the n -gap, and possible prior appearance of the trick).

One honest qualification: "determine the correct dependence" read maximally strictly (e.g., pin down $\lim ES_l(n)/l$ for each n , or the exact function) is not achieved; what is achieved is that the dependence is $\Theta(l)$ uniformly. The writeup's verdict line says exactly this and does not overclaim.

B.2 Step-by-step findings

Steps S1-S11 are the alteration Lemma (Section 1 of the writeup), S12 is the transfer inequality (Section 2), S13-S17 are the application (Section 3). Every computation and inequality was re-derived by hand and re-checked by the scripts described below.

step	label	note
S1. Lemma statement: P an N-set with $< l$ points on every line contains Q with no k collinear, $ Q \geq (1/4)(N/l)^{\{(k-2)/(k-1)\}}$	VALID	Proved by S2-S11; also stress-tested empirically on 30 concrete instances (grids, unions of lines, clusters, random sets), always satisfied with large margin.
S2. Identity $C(s,k) = C(s,2) C(s-2,k-2) / C(k,2)$	VALID	Exact identity; verified symbolically (Fraction arithmetic) for all $3 \leq k \leq 11, k \leq s < 60$ (check A1).
S3. $C(s-2,k-2)/C(k,2) \leq l^{\{k-2\}}$ for $s \leq l-1$	VALID	$C(s-2,k-2) \leq (s-2)^{\{k-2\}} \leq l^{\{k-2\}}$ and $C(k,2) \geq 1$; verified for $k \leq 11, l \leq 39$ (check A2). Degenerate $l=3$ case: no line has $\geq k \geq 3$ points, $e = 0$, vacuous.
S4. $\sum_L C(s_L,2) \leq C(N,2)$	VALID	Each pair of distinct points spans a unique line, so the sums of $C(s_L,2)$ over all lines counts each pair at most once.
S5. Inequality (1): $e \leq l^{\{k-2\}}$ $\sum_L C(s_L,2) < N^2 l^{\{k-2\}}$	VALID	Follows from S2-S4 and $C(N,2) < N^2/2 < N^2$. Strictness fine for $N \geq 2$ (and $e = 0 \leq$ anything for $N < 2$). Empirically confirmed on all 30 test instances.

step	label	note
S6. Case $e < N/2$: deleting one vertex per edge leaves an independent set $> N/2 \geq (1/4)(N/l)^{\{(k-2)/(k-1)\}}$	VALID	$N - e > N/2$; and $(N/l)^a \leq N^a \leq N$ for $a = (k-2)/(k-1)$ in $(0,1)$, $N \geq 1$, $l \geq 1$, so $(1/4)(N/l)^a \leq N/4 < N/2$ (check D).
S7. Case $e \geq N/2$: $p = (N/2e)^{\{1/(k-1)\}} \leq 1$	VALID	Immediate from $e \geq N/2$.
S8. $E[X - Y] = pN - p^k e = pN/2$	VALID	$p^k e = p * p^{\{k-1\}} e = p * (N/2e) * e = pN/2$. Verified symbolically in sympy for $k = 3..14$ (check B).
S9. Some outcome has $X - Y \geq pN/2$; deleting one vertex from each surviving edge leaves an independent set of size $\geq pN/2$	VALID	Standard first-moment alteration; deletion removes at most Y vertices from the X chosen ones and destroys every contained edge.
S10. $pN/2 \geq (N/2)(1/(2N l^{\{k-2\}})\{1/(k-1)\} = 2^{\{-1-1/(k-1)\}} (N/l)^{\{(k-2)/(k-1)\}} \geq (1/4)(N/l)^{\{(k-2)/(k-1)\}}$	VALID	First inequality from (1); the algebraic identity verified symbolically for $k = 3..14$ (check C1); $2^{\{-1-1/(k-1)\}} \geq 1/4$ iff $1/(k-1) \leq 1$, true for $k \geq 2$, verified for k up to 1000 (check C2). Worst case $k = 3$ gives $2^{\{-3/2\}} \sim 0.354 > 0.25$.
S11. Independent set in $H =$ subset with no k collinear points	VALID	Edges of H are exactly the collinear k -subsets of P ; any k collinear points of an independent set would form an edge.
S12. Transfer inequality: with $M = ES_k(n)$, $r = (k-1)/(k-2)$, $N = \text{ceil}(l (4M)^r)$, every N -set contains l collinear or n in convex position	VALID	If no l collinear, S1 gives Q with no k collinear and $ Q \geq (1/4)(N/l)^{\{1/r\}} \geq (1/4)(4M)^r = M$ (since $N/l \geq (4M)^r$ exactly); any M -point subset of Q contains k collinear (impossible in Q) or n in convex position. Grid-verified for $k = 3..14$, M and l over wide ranges (check E). Sanity check at $k = n = 3$: $ES_3(3) = 3$ (any non-collinear triple is in convex position), the transfer gives $ES_l(3) \leq 144 l$, consistent with the true value $ES_l(3) = 1$.
S13. Instantiation (2): $ES_l(n) \leq \text{ceil}(l (4 ES_n(n))^{\{(n-1)/(n-2)\}})$	VALID	$k = n \geq 3$ is a legal choice in S12.

step	label	note
S14. Bound (3): $ES_n(n) < n^2 2^{\{n + C_0 \sqrt{n \log n}\}}$	VALID	The source paper's Theorem 1.1 holds "for each $l, n \geq 3$ " (verified in the TeX source), so setting $l = n$ is legitimate; this is the diagonal instance, not a circular use (the transfer consumes it as a black box).
S15. Logarithmic estimate: $r \log_2(4M) \leq n + C_1 \sqrt{n \log n}$, ceiling absorbed	VALID	$r(n + 2 \log_2 n + 2 + C_0' \sqrt{n \log_2 n}) - n \leq C_1 \sqrt{n \log_2 n}$ for an absolute C_1 : verified numerically for all n in $[3, 20000]$ plus $10^5, 10^6$; e.g., $C_0' = 3$ needs $C_1 \sim 12.18$ (worst case at $n = 3$), including the ceiling factor $\log_2(13/12)$ (the argument of the ceiling is ≥ 12 , so $\text{ceil}(x) \leq (13/12)x$) (check F). Every term beyond n is indeed $O(\sqrt{n \log n})$ with an absolute constant since $\sqrt{n \log_2 n} \geq \sqrt{3 \log_2 3} > 2$ for $n \geq 3$.
S16. Lower-bound simplification: $(3l-1) 2^{\{n-5\}} < ES_l(n)$ implies $ES_l(n) > (1/12) l 2^n$ for $l \geq 3$	VALID	$(3l-1) 2^{\{n-5\}} - (1/12) l 2^n = 2^n (l-3)/96 \geq 0$ (sympy factorization, check G), with equality only at $l = 3$; strictness then follows from the strict inequality $ES_l(n) > (3l-1) 2^{\{n-5\}}$ (the source paper's Theorem 1.2 actually gives $ES_l(n) \geq (3l-1) 2^{\{n-5\}} + 1$).
S17. Final display $(1/12) l 2^n < ES_l(n) \leq l 2^{\{n + C_1 \sqrt{n \log n}\}}$ and the conclusion of uniformly linear dependence on l	VALID	Direct combination of S13-S16.

No step is a GAP or ERROR. The writeup's "Remaining gap" section is accurate: the $2^{O(\sqrt{n \log n})}$ factor in n is not closed (nor claimed).

B.3 Reference check

All external inputs were fetched and verified against primary sources:

1. **arXiv:2405.03455 (source paper)**. I downloaded the TeX source (arxiv.org/e-print/2405.03455, saved under the scratchpad) and read it in full. Verified: (a) the definition of $ES_l(n)$ matches the writeup's usage;

- (b) Theorem 1.1: “There exists $C > 0$ such that, for each $l, n \geq 3$, $ES_l(n) \leq l^2 \cdot 2^{n+C \sqrt{n \log n}}$ ” — exactly the hypothesis range ($l, n \geq 3$) the writeup needs for the diagonal instance $l = n$; (c) Theorem 1.2: “For each $l, n \geq 3$, $ES_l(n) \geq (3l-1) \cdot 2^{n-5} + 1$ ” — supplying the strict lower bound the writeup quotes; (d) the open-problem sentence appears verbatim. The paper’s own upper-bound proof (positive-fraction cups-caps + simplicial partitions) is entirely different from the writeup’s argument and produces the l^2 via a $(l+n)^2$ factor per level; nothing in the paper anticipates the bootstrap. The catalog notes the paper was published in Computational Geometry 131 (2026), art. 102218, without revising the open problem.
2. **Novelty scan (citing literature).** Semantic Scholar lists exactly one paper citing 2405.03455: Koki Furukawa, “Big convex polytopes or rich hyperplanes”, arXiv:2501.03645 (2025). I downloaded its TeX source and read the relevant sections. Furukawa proves, via a maximal-general-position-subset argument, $ES_2(l,n) \leq (l-3) C(ES(n)-1, 2) + ES(n) \sim l \cdot 4^{n+O(\sqrt{n \log n})}$ and explicitly writes: “However, this bound is weaker than the one given by Conlon et al. unless l is significantly larger than n .” So the literature’s best known bounds remain $l^2 \cdot 2^{n+O(\sqrt{n \log n})}$ (small l) and $l \cdot 4^{n+...}$ (huge l); the writeup’s uniform $l \cdot 2^{n+O(\sqrt{n \log n})}$ appears in neither and is strictly stronger than both. The catalog’s auto-review (2026-05, three targeted searches) also found no resolution; my own searches (Erdos-Szekeres collinear linear dependence; general position subset selection) found nothing closer.
 3. **Technique provenance.** The Lemma is a standard k -uniform-hypergraph alteration (Spencer-type independence bound); its $k = 3$ case is the classical step in the general position subset selection literature (Payne-Wood, SIAM J. Discrete Math. 2013, arXiv:1208.5289, use exactly this kind of extraction, with sharper tools). This supports correctness; the specific bootstrap through the diagonal value $ES_n(n)$ — which is what makes the exponent loss $(k-1)/(k-2)$ harmless — is not something I could find anywhere.

No misquoted or nonexistent citation was found; every quoted bound matches its source with identical hypotheses.

B.4 Computational check

Scripts (reproducible) in `verification/scripts/2405.03455__00/`, run with Python 3.14 + sympy 1.14:

- `check_inequalities.py` — all algebraic/analytic steps. Results: the binomial identity holds exactly for $3 \leq k \leq 11$, $k \leq s < 60$ (A1); $C(s,k) \leq C(s,2) l^{k-2}$ for $s \leq l-1$ (A2); $E[X-Y] = pN/2$ symbolically for $k = 3..14$ (B); the S10 chain identity symbolically for $k = 3..14$ (C1); $2^{\lfloor 1-1/(k-1) \rfloor} \geq 1/4$ for all k in $[3,1000]$ (C2); the $e < N/2$ case inequality (D); the transfer arithmetic $(1/4)(N/l)^{1/r} \geq M$ on a grid of k in $3..14$, M in $\{3,...,12345\}$, l in $\{3,...,10^6\}$ (E); the Section 3 asymptotics for all n in $[3, 20000]$ and $n = 10^5, 10^6$, including the ceiling factor — e.g., $C_0' = 1, 3, 10$ need $C_1 = 8.18, 12.18, 26.18$ respectively, worst case at $n = 3$ (F); and $(3l-1)2^{n-5} - (1/12) l^2 n = 2^n (l-3)/96 \geq 0$ exactly (G). ALL PASS.
- `check_lemma_empirical.py` — stress-test of the Lemma itself on 30 concrete instances: $5 \times 5/8 \times 8/12 \times 12$ integer grids, unions of $3/4/10$ long lines, 12 collinear clusters of 6 points (the lower-bound-construction shape), and random integer sets, for k from 3 up to the maximal collinearity. In every instance the edge bound (1) holds (e.g., 12×12 grid, $k = 3$: $e = 10332 \leq 133848$) and a greedy deletion already exhibits a no- k -collinear subset far above the Lemma’s guarantee (e.g., 3 lines $\times$ 40 points, $l = 41$, $k = 3$: found 6 ≥ 0.43 required; 12×12 grid, $k = 4$: found 27 ≥ 1.24 required). ALL PASS.

The writeup rests on no single explicit finite object; the finite substance is the inequality chain, all of which was machine-checked. No claimed property failed.

B.5 Caveats

1. **Conditional on the source paper.** The result consumes Theorem 1.1 of arXiv:2405.03455 (at $l = n$) and Theorem 1.2 as black boxes. Both are peer-reviewed (Comput. Geom. 131 (2026)); I did not independently referee the source paper’s own proofs, but its Section 2-4 arguments read plausibly and the theorem statements are used exactly as published.
2. **Novelty is surprising and only negatively certified.** A six-author team of leading combinatorialists left the $l^2 \rightarrow l$ question open, and Furukawa (2025) revisited the l -dependence without finding this two-page bootstrap; my citation sweep found only that one citing paper. I cannot exclude that the observation is folklore or sitting in an unindexed preprint; the writeup itself flags exactly this. This affects novelty, not correctness.
3. **What “determine the correct dependence” now means.** The l -dependence is pinned to $\Theta(l)$ uniformly; the ratio of the two bounds is $12 * 2^{\{C_1 \sqrt{n \log n}\}}$, a function of n alone, and matching the classical $ES(n)$ gap. If one insists that “correct dependence on l ” includes the exact n -dependent coefficient of l , that remains open — but that reading would make the problem subsume the classical Erdos-Szekeres conjecture.
4. **Conventions.** “ l collinear” means at least l (both paper and writeup); n points in convex position means vertices of a convex n -gon (so collinear n -tuples do not count); the writeup is consistent throughout. The equality case $l = 3$ in the final lower-bound display needs the “+1” strictness of the paper’s Theorem 1.2 (or the strict “ $<$ ” of its abstract), which holds.
5. **Edge cases.** $l = 3$ or small N make the hypergraph empty and the Lemma vacuous-but-true; $k = n = 3$ gives $ES_l(3) \leq 144 l$ against the true value l (wasteful but correct); all constants were checked at the worst case $n = 3, k = 3$. No degenerate case breaks any step.
6. The writeup’s headline constant $1/12$ in the lower display is tight only at $l = 3$ and is inherited from the source construction, not new.

B.6 Referee summary

I set out assuming the writeup was wrong and could not break it. The argument is a short, fully elementary bootstrap: a textbook alteration lemma extracts, from any N -point set with fewer than l collinear, a subset of size $(1/4)(N/l)^{\{(k-2)/(k-1)\}}$ with no k collinear; choosing $k = n$ and feeding the subset into the source paper’s own diagonal bound $ES_n(n) < n^2 2^{\{n+C \sqrt{n \log n}\}}$ makes the exponent loss $(n-1)/(n-2)$ cost only a constant-plus-subexponential factor, yielding $ES_l(n) \leq l * 2^{\{n+C_1 \sqrt{n \log n}\}}$ for all $l, n \geq 3$. Every identity, inequality, and asymptotic absorption was re-derived and machine-verified; the source paper’s theorems were checked verbatim in the arXiv TeX; the only citing paper (Furukawa, arXiv:2501.03645) proves a strictly weaker planar bound and explicitly acknowledges the regime where it loses, confirming the claimed bound was not in the indexed literature. The interpretation is the intended one, and the writeup honestly scopes what remains open (the n -gap common to classical Erdos-Szekeres). Verdict: CONFIRMED, with the caveat that the result’s simplicity relative to the expertise of the source authors makes independent expert scrutiny and a folklore check advisable before publication.

References

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- [3] M. S. Payne and D. R. Wood, On the general position subset selection problem, *SIAM J. Discrete Math.* 27 (2013) 1727–1733; arXiv:1208.5289.